Assignment 8: Basic SQL Queries and Aggregate Functions

Filtering, sorting, grouping and summaries

Project Topic: Hotel Management System

1. Objective

This assignment demonstrates filtering, sorting, grouping and aggregate functions using the sample hotel data from Assignment 7.

2. WHERE: Find confirmed reservations

```
SELECT reservation_id, guest_id, booking_status\nFROM Reservations\nWHERE booking_status = 'Confirmed';
```

Expected output: reservation IDs 1004 and 1005. WHERE filters individual rows before they are returned.

3. ORDER BY: Sort room prices from highest to lowest

```
SELECT type_name, base_price\nFROM RoomTypes\nORDER BY base_price DESC;
```

Expected output: Suite 6500.00; Deluxe 3200.00; Single 1800.00. ORDER BY sorts the result.

4. GROUP BY: Count reservations by status

```
SELECT booking_status, COUNT(*) AS reservation_count\nFROM Reservations\nGROUP BY booking_status\nORDER BY booking_status;
```

booking_status	reservation_count
Checked-In	2
Checked-Out	1
Confirmed	2

GROUP BY creates one summary row for each status.

5. HAVING: Find statuses with more than one reservation

```
SELECT booking_status, COUNT(*) AS reservation_count\nFROM Reservations\nGROUP BY booking_status\nHAVING COUNT(*) > 1;
```

booking_status	reservation_count
Checked-In	2
Confirmed	2

HAVING filters groups after GROUP BY; it is used for conditions on aggregate results.

6. Aggregate Functions

COUNT(): Count all reservations

```
SELECT COUNT(*) AS total_reservations FROM Reservations;
```

Expected output: 5.

SUM(): Total amount of payment records

```
SELECT SUM(amount) AS total_payment_amount FROM Payments;
```

Expected output: 27600.00. This is the sum of all recorded payment amounts, including the payment marked Pending; it should not be interpreted as money fully collected.

AVG(): Average standard room price

```
SELECT AVG(base_price) AS average_room_price FROM RoomTypes;
```

Expected output: 3833.33 (rounded to two decimal places).

MIN(): Lowest room price

```
SELECT MIN(base_price) AS lowest_room_price FROM RoomTypes;
```

Expected output: 1800.00.

MAX(): Highest room price

```
SELECT MAX(base_price) AS highest_room_price FROM RoomTypes;
```

Expected output: 6500.00.

7. Additional Filtering and Sorting

```
SELECT reservation_id, check_in_date, check_out_date\nFROM Reservations\nWHERE check_in_date >= '2026-10-15'\nORDER BY check_
```

This returns reservations whose check-in date is on or after 15 October 2026, sorted from earliest to latest.

8. Conclusion

The queries demonstrate WHERE, ORDER BY, GROUP BY, HAVING and COUNT, SUM, AVG, MIN and MAX. The output examples are based on the sample records in Assignment 7.